

Digital ID for Africa Hackathon: Ideation Phase Submission

Team Information

Team Name: MedAxis Africa

Team Members (4):

- Ilyass Lhafi - Male - Al Akhawayn University in Ifrane – i.lhafi@au.ma
- Lina Lassri - Female - Al Akhawayn University in Ifrane – l.lassri@au.ma / linalassri2006@gmail.com
- Youssef Assemblali - Male - Al Akhawayn University in Ifrane – y.assemblali@au.ma / youssefsseml@gmail.com
- Salma Essagar – Female – Al Akhawayn University in Ifrane – s.essagar@au.ma / essagarsalma@gmail.com

University/Institution:

Al Akhawayn University, Ifrane, Morocco

Patron:

Houda Chakiri – Assistant Professor of Computer Science, School of Science and Engineering, Al Akhawayn University – H.Chakiri@au.ma – +212 535 86 3161

Idea Submission

1. Sector:

Healthcare – National Digital Prescription Infrastructure

2. Title:

MedAxis: Digital ID–Anchored National Prescription Trust Infrastructure.

3. Problem Statement

In Morocco and across Africa, paper-based prescriptions enable forgery and misuse. Pharmacies cannot instantly verify practitioner licensing, patient identity, or prevent duplicate dispensing (reusing prescriptions). Without a unified national audit trail, drug abuse, over-prescription, and illegal practices spread, causing drug leakage and eroding public trust. This creates massive inefficiencies for doctors, pharmacists, and regulators. Additionally, paper-dependent insurance processes force patients to pay upfront and submit manual claims. This burden leads to reimbursement delays and high risks of fraud, further straining the healthcare ecosystem and its participants.

4. Proposed Solution

MedAxis is a digital prescription infrastructure anchored in National Digital Identity systems (e.g., CNIE in Morocco or MOSIP-based systems in other African countries). It does not create or manage its own identity database. All practitioners and patient verification is delegated to the country's official Digital ID authentication service (such as a MOSIP gateway or CNIE-based verification system). The platform receives only a secure verification token confirming identity and does not access or store biometric or core identity records. MedAxis does not replace hospital systems or centralize medical records; it establishes a secure national prescription trust layer.

A. Licensed Practitioner Authentication

Doctors and pharmacists authenticate using their National Digital ID. Only verified, licensed professionals can issue or dispense prescriptions, preventing fraudulent or unlicensed practice.

B. Secure Digital Prescription Issuance

Each prescription generates a unique, digitally signed prescription ID. The system stores minimal metadata in a National Prescription Ledger:

- Prescription ID
- Verified doctor ID
- Tokenized patient ID
- Drug code and dosage
- Timestamp
- Expiry date

No diagnoses, clinical notes, or full medical records are stored centrally.

C. Pharmacy Verification and Dispensing

At the pharmacy, the pharmacist authenticates via National Digital ID, and the patient's identity is verified through the official Digital ID authentication service (e.g., OTP or assisted verification where supported). The platform proceeds only after identity confirmation from the national ID system. It verifies that:

- The doctor is licensed
- The prescription is valid and not expired
- The medication has not already been dispensed

Upon dispensing, the pharmacist logs date and quantity, preventing duplication, reuse, and pharmacy hopping.

D. Integrated Insurance Verification and Real-Time Co-Payment

Through secure APIs or phased integration with insurance providers, the system determines coverage eligibility and reimbursement percentage. Instead of paying the full amount and filing paper claims, the patient pays only their co-payment at the pharmacy. The insurance portion is recorded digitally, reducing fraud and reimbursement delays.

E. Patient Read-Only Access Layer

Patients do not issue prescriptions themselves; doctors link prescriptions to the patient's verified identity during consultation. A secure read-only view allows patients to see minimal prescription history (medication, dosage, issuing facility, dispensing status) and report discrepancies, improving transparency without exposing sensitive data.

F. Privacy-Respecting Hybrid Architecture

Clinical records remain within hospitals and private practices. Identity verification requests are routed through the national Digital ID gateway, ensuring MedAxis never stores biometric data. Only prescription metadata is centralized. All transactions are encrypted, digitally signed, and governed by role-based access control to ensure security, integrity, and auditability.

MedAxis transforms prescriptions from paper-based artifacts into verifiable digital assets anchored in national identity.

5. Novelty

MedAxis differs from traditional electronic medical record systems by focusing exclusively on prescription authenticity and trust infrastructure rather than full medical record centralization. Instead of digitizing entire patient histories, it establishes a national, identity-anchored prescription ledger that ensures every prescription is verifiable, traceable, and issued by a licensed practitioner. By combining Digital ID authentication for medical staff with secure prescription issuance and pharmacy validation, the system addresses prescription fraud, drug abuse, and unlicensed practice in a unified manner. The addition of a privacy-preserving, read-only patient transparency layer further distinguishes it from existing siloed e-health solutions.

6. Feasibility

In Morocco, MedAxis can be piloted initially within public hospitals for controlled-substance prescriptions (e.g., opioids, psychiatric medications, and high-risk antibiotics) using CNIE-based practitioner authentication. The system integrates via secure APIs or phased data exchange with Ministry of Health registries if the law allows. It does not require replacing existing hospital systems, reducing cost and resistance. The hybrid architecture minimizes legal risk by storing only prescription metadata, aligning with data protection principles. Adoption is encouraged through regulatory alignment with pharmacy and practitioner licensing frameworks and for controlled substances, regulatory mandates can require digital issuance and verification, accelerating national adoption. Technically, the platform can be prototyped using MOSIP sandbox authentication services, demonstrating integration with national identity infrastructure without replicating identity databases.

7. Impact

MedAxis reduces prescription fraud, prevents drug abuse, and strengthens regulatory oversight while protecting patient privacy. It improves patient experience by eliminating paper-based reimbursement processes and reducing waiting times at pharmacies. By creating a unified national prescription audit trail, MedAxis ensures that only licensed practitioners issue prescriptions and that every dispensing event is logged. Scaled across Africa, the infrastructure can combat antimicrobial resistance, reduce controlled-drug leakage, provide authorities with clearer visibility over emerging health risks, and strengthen pharmaceutical governance. It aligns with Digital Public Infrastructure goals by using national identity as a secure trust anchor in healthcare.

Originality Statement

We, the undersigned team members, confirm that the idea submitted in this document is our original work and has not been previously published or submitted elsewhere to the best of our knowledge.

Team Member Signatures

– Ilyass Lhafi

– Lina Lassri



– Youssef Assemblali

– Salma Essagar

References

- World Health Organization. “Antimicrobial resistance.”
- Ministry of Health Morocco – Regulatory framework for licensed practitioners and pharmacies.
- MOSIP Documentation – Digital Identity Authentication Framework.